Why temperature only rescales time: the model in dimensionless form

enceladus_DSM `studies/keff_sintering/temperature_scaling/`

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Result. For isothermal sintering by vapor transport, the model reduces to a problem with a single parameter, $\Lambda = L^*/\bar{R}$, once time is measured in units of $\tau = \bar{R}^2\beta_{\mathrm{sub}}/d_0$. The saturation vapor density, which changes 31-fold between -40 and $-5\,^{\circ}\mathrm{C}$, is entirely inside τ. Λ contains the vapor diffusivity and changes by 20%. Numbers are from `check_groups.py` (output `groups.txt`).

1 The dimensional problem

The phase-field model is used in its sharp-interface limit, which is the limit it is built to reproduce (Kaempfer & Plapp, 2009). The aggregate runs are isothermal, so temperature is a constant parameter T. Three statements remain.

Vapor in the pore. From Eq. (1c) of the manuscript, away from the interface,

$$\frac{\partial \rho_v}{\partial t} = D_v \nabla^2 \rho_v. \tag{1}$$

Mass balance at the interface. Let $\mathbf{n}$ be the unit normal pointing from the ice into the pore and v_n the velocity of the interface along $\mathbf{n}$ ($v_n > 0$ when ice grows). Ice that grows takes its mass from the vapor arriving by diffusion. Integrating Eq. (1c) across the interface, and neglecting ρ_v against ρ_{ice} (a ratio of 10^{-6}),

$$\rho_{\mathrm{ice}}\, v_n = D_v\, \frac{\partial \rho_v}{\partial n}. \tag{2}$$

The sign: above a growing surface the vapor is depleted, so ρ_v rises with distance from the ice and $\partial\rho_v/\partial n > 0$.

Gibbs–Thomson condition at the interface (manuscript Eq. 6),

$$\frac{\rho_v - \rho_{vs}^I}{\rho_{vs}^I} = d_0\,\chi + \beta\, v_n, \qquad \beta = \frac{1}{\alpha_c}\,\frac{\rho_{\mathrm{ice}}}{\rho_{vs}^I}\sqrt{\frac{2\pi m}{k_B T}}, \tag{3}$$

with χ the curvature (positive for a convex grain). Here β stands for $\beta_{\mathrm{sub}} + \Delta\beta$; the thin-interface term is 2% of β_{sub} at every temperature in the aggregates (Section 6), so it rescales β by a constant. The temperature-dependent quantities are $\rho_{vs}^I(T)$, $D_v(T)$, $d_0(T) \propto 1/T$ and the explicit $\sqrt{T}$.

2 Scales

Quantity	Scale	Reason
length, $\mathbf{x} = \bar{R}\,\mathbf{x}'$	$\bar{R}$	mean grain radius
curvature, $\chi = \chi'/\bar{R}$	$1/\bar{R}$	follows from the length
supersaturation, $\dfrac{\rho_v - \rho_{vs}^I}{\rho_{vs}^I} = \dfrac{d_0}{\bar{R}}\,u$	$d_0/\bar{R}$	what curvature $1/\bar{R}$ produces
velocity, $v_n = V\,v'$	$V = \dfrac{d_0}{\bar{R}\,\beta}$	makes the kinetic term of order one
time, $t = \tau\,t'$	$\tau = \dfrac{\bar{R}}{V} = \dfrac{\bar{R}^2\beta}{d_0}$	time to move one grain radius

So u is the supersaturation measured in units of the equilibrium supersaturation of a mean grain, and $\rho_v = \rho_{vs}^I\,(1 + (d_0/\bar{R})\,u)$.

3 Substitution

Gibbs–Thomson, Eq. (3):

$$\frac{d_0}{\bar{R}}\,u = d_0\,\frac{\chi'}{\bar{R}} + \beta\,\frac{d_0}{\bar{R}\,\beta}\,v' \quad\Longrightarrow\quad \boxed{u = \chi' + v'}$$

No parameter is left.

Mass balance, Eq. (2). The left side is $\rho_{\text{ice}} V v'$. On the right, $\partial\rho_v/\partial n = \rho_{vs}^I\,(d_0/\bar{R})\,(1/\bar{R})\,\partial u/\partial n'$. Therefore

$$v' = \frac{D_v\,\rho_{vs}^I\,d_0}{\rho_{\text{ice}}\,V\,\bar{R}^2}\,\frac{\partial u}{\partial n'} = \frac{D_v\,\rho_{vs}^I\,\beta}{\rho_{\text{ice}}\,\bar{R}}\,\frac{\partial u}{\partial n'},$$

using $V = d_0/(\bar{R}\beta)$. Define

$$L^* \equiv \frac{D_v\,\beta\,\rho_{vs}^I}{\rho_{\text{ice}}} = \frac{D_v(T)}{\alpha_c}\sqrt{\frac{2\pi m}{k_B T}}, \qquad \Lambda \equiv \frac{L^*}{\bar{R}}, \tag{4}$$

where the second form of L^* follows from the definition of β in Eq. (3): **the saturation vapor density cancels.** Then

$$\boxed{v' = \Lambda\,\frac{\partial u}{\partial n'}}$$

Diffusion, Eq. (1). Both sides carry the factor $\rho_{vs}^I d_0/\bar{R}$:

$$\frac{1}{\tau}\frac{\partial u}{\partial t'} = \frac{D_v}{\bar{R}^2}\nabla'^2 u \quad\Longrightarrow\quad P\,\frac{\partial u}{\partial t'} = \nabla'^2 u, \qquad P \equiv \frac{\bar{R}^2}{D_v\,\tau} = \frac{d_0}{D_v\,\beta}.$$

P is the time for vapor to diffuse across a grain divided by the time for the grain to change. It is 10^{-11} (Section 6), so the vapor field is always in equilibrium with the current geometry:

$$\boxed{\nabla'^2 u = 0}$$

4 The dimensionless problem

$$\nabla'^2 u = 0 \ \text{ in the pore;} \qquad u = \chi' + v', \quad v' = \Lambda \frac{\partial u}{\partial n'} \ \text{ on the ice surface,} \tag{5}$$

with periodic boundaries. Eliminating v' gives a single boundary condition, $u - \Lambda\, \partial u/\partial n' = \chi'$.

Given a starting geometry, the evolution in $t' = t/\tau$ depends on Λ and on nothing else. Two limits make its meaning clear:

- $\Lambda \to \infty$: u is uniform, attachment is the slow step, and each surface moves at $v' = \bar{u} - \chi'$ with $\bar{u}$ set by conservation of ice. The evolution in t/τ is then identical at every temperature.
- $\Lambda \to 0$: $u = \chi'$ on the surface and diffusion is the slow step.

The aggregates have $\Lambda = 2.5$ to 2.9: attachment is the larger resistance over distances shorter than $L^* \approx 135$ μm, which covers transport between neighboring surfaces, but diffusion is not negligible.

5 Where temperature enters

	$-40\,^\circ$C	$-30\,^\circ$C	$-20\,^\circ$C	$-10\,^\circ$C	$-5\,^\circ$C
ρ^I_{vs} [10^{-4} kg m^{-3}]	1.055	3.123	8.487	21.39	33.11
τ [days]	1713	591	222	89.8	58.5
$\tau/\tau(-5\,^\circ\text{C})$	29.3	10.1	3.79	1.53	1
L^* [μm]	122.5	129.4	136.5	143.6	147.1
$\Lambda = L^*/\bar{R}$	2.45	2.59	2.73	2.87	2.94

- **The time scale** carries a factor of 29.3 across the range: $\tau \propto \beta/d_0 \propto \sqrt{T}/\rho^I_{vs}(T)$ (since $d_0 \propto 1/T$ and $\beta \propto 1/(\rho^I_{vs}\sqrt{T})$). The factor is 31.4 from ρ^I_{vs}, times 1.072 from $\sqrt{T}$, divided by 1.150 from d_0.
- **The one remaining group** changes by 20%: $L^* \propto D_v/\sqrt{T} \propto T^{1.31}$.

So both temperature-dependent transport properties are accounted for. The saturation vapor density sets the clock. The diffusivity survives only in Λ, weakly, and its effect is further reduced because attachment is the larger resistance. This is why curves at different temperatures superpose when plotted against t/τ, closely but not exactly: the simulated rates follow τ_{sub} to within 4%. A 20% change in Λ is the right size to produce that; it has not been tested as the cause.

Relation to τ_{sub}. The manuscript's clock is $\tau_{\text{sub}} = \varepsilon^2(\beta_{\text{sub}} + \Delta\beta)/d_0 = (\varepsilon/\bar{R})^2\,\tau$ with β including $\Delta\beta$. With $\varepsilon = \bar{R}/50$ fixed, the two differ by the constant 2500, so ratios between temperatures are the same. Check at $-20\,^\circ$C: $(1/50)^2 \times 221.9$ d $\times\, 1.020 = 7822$ s, the value in Table 2.

6 What was neglected, with sizes

Neglected	Group	Size, -40 to $-5\,^{\circ}\mathrm{C}$
Vapor storage (time derivative in Eq. 1)	$P/\xi_v = d_0/(\xi_v D_v \beta)$	10^{-9} to 2×10^{-8}
Latent heat changing the local saturation	$G_T = \dfrac{\rho_{\mathrm{ice}} L_{\mathrm{sub}}^2 m \bar{R}}{k_B T^2 k_{\mathrm{ice}} \beta}$	10^{-4} to 2.5×10^{-3}
ρ_v against ρ_{ice} in Eq. (2)	$\rho_{vs}^I/\rho_{\mathrm{ice}}$	10^{-7} to 4×10^{-6}
Thin-interface term, as a change with T	$\Delta\beta/\beta_{\mathrm{sub}}$	0.020 at every T

- *Storage.* The solver multiplies the diffusion and source terms of the vapor equation by $\xi_v = 10^{-3}$ together, which slows the vapor's relaxation by 10^3 and leaves the balance (2) unchanged. The storage group becomes P/ξ_v, still negligible.
- *Latent heat.* Growth at velocity V releases heat that changes the surface temperature by about $\rho_{\mathrm{ice}} L_{\mathrm{sub}} V \bar{R}/k_{\mathrm{ice}} \sim 10^{-7}$ K, which shifts the local saturation by G_T in units of u. This group does depend strongly on temperature (it contains ρ_{vs}^I), but it is at most 2.5×10^{-3}. It is the only place the heat equation enters.
- *Thin-interface term.* $\Delta\beta$ and β_{sub} are both proportional to $\rho_{\mathrm{ice}}/\rho_{vs}^I$, so their ratio is nearly independent of temperature and β is rescaled by a constant 1.020.

7 What this does and does not show

- It shows that the collapse in t/τ_{sub} follows from the structure of the model, for any geometry, as long as α_c does not depend on temperature. It is a property of the model.
- It identifies what would break the collapse: an $\alpha_c(T)$ (it would enter Λ as well as τ); a second mechanism with its own temperature dependence; a temperature gradient, which makes ρ_{vs}^I vary in space; or a much larger α_c or grain size, which lowers Λ and makes the diffusivity matter.
- It is a sharp-interface argument. The simulations add the check that the diffuse-interface model behaves this way: k_{eff} against SSA agrees across temperature to 0.27%, rates to 4%.